

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) marketing_..._users.csv

marketing_ab_test_12k_users.csv(text/csv) - 205857 bytes, last modified: 2/2/2026 - 100% done

Saving marketing_ab_test_12k_users.csv to marketing_ab_test_12k_users.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
```

```
df = pd.read_csv("/content/marketing_ab_test_12k_users.csv")
```

```
# Assume columns: group (control/test), converted (0/1), revenue(optional)
df.head()
```

	user_id	group	converted	revenue	
0	1	control	0	0.0	
1	2	test	0	0.0	
2	3	test	0	0.0	
3	4	test	0	0.0	
4	5	control	0	0.0	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
# Split groups
control = df[df['group'] == 'control']
test = df[df['group'] == 'test']
```

Double-click (or enter) to edit

```
alpha = 0.05
```

```
#Calculate Group Metrics
control_rate = control['converted'].mean()
test_rate = test['converted'].mean()

control_rate, test_rate
```

```
(np.float64(0.08403361344537816), np.float64(0.11532625189681335))
```

```
#Run Test & Extract p-value
from scipy.stats import chi2_contingency

contingency = pd.crosstab(df['group'], df['converted'])
chi2, p_value, dof, expected = chi2_contingency(contingency)

p_value
```

```
np.float64(1.2330184558392093e-08)
```

```
if p_value < alpha:
    result = "Statistically Significant"
else:
    result = "Not Statistically Significant"

result
```

```
'Statistically Significant'
```

```
from statsmodels.stats.proportion import proportion_confint

n_control = len(control)
n_test = len(test)

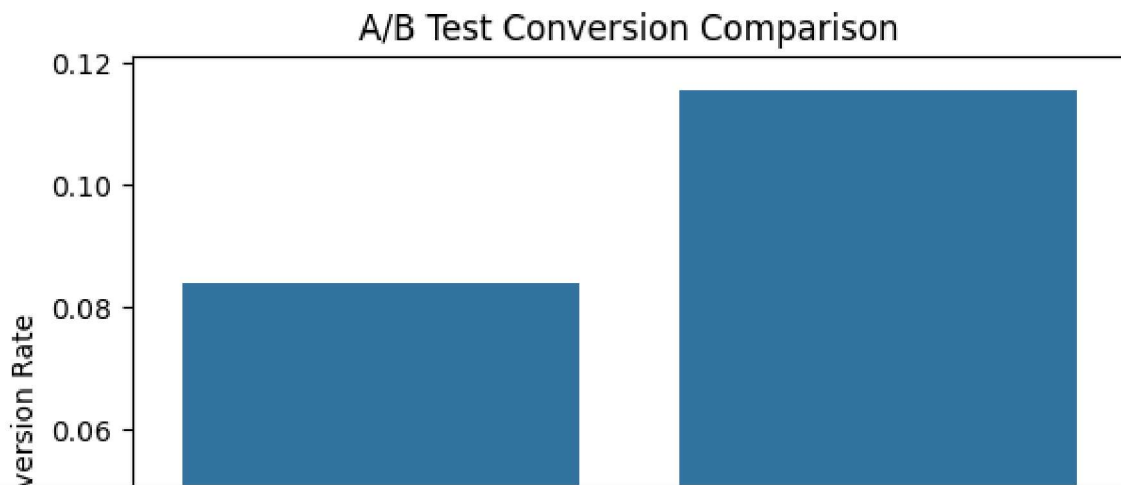
ci_control = proportion_confint(control['converted'].sum(), n_control, alpha=0.05)
ci_test = proportion_confint(test['converted'].sum(), n_test, alpha=0.05)

ci_control, ci_test
```

```
((0.07705360722180547, 0.09101361966895084),
 (0.107197202025626, 0.1234553017680007))
```

```
#Calculate Group Distributions
```

```
#visualize Group Distributions
sns.barplot(x=['Control', 'Test'], y=
plt.ylabel("Conversion Rate")
plt.title("A/B Test Conversion Compari
plt.show()
```



```
#Final Decision & Business Recommendation
summary = pd.DataFrame({
    "Group": ["Control", "Test"],
    "Conversion Rate": [control_rate, test_rate]
})

summary.to_csv("ab_test_summary.csv")
summary
```

	Group	Conversion Rate	
0	Control	0.084034	
1	Test	0.115326	

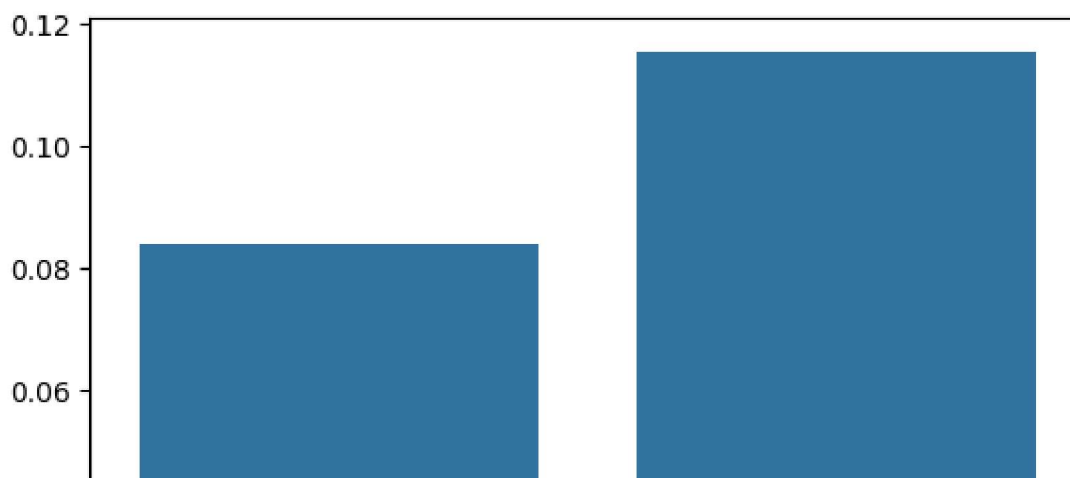


Next steps:

[Generate code with summary](#)[New interactive sheet](#)

```
sns.barplot(x=['Control', 'Test'], y=[control_rate, test_rate])
```

<Axes: >



```
contingency = pd.crosstab(df['group']
chi2, p_value, dof, expected = chi2_
```

```
!pip install reportlab
```

```
Collecting reportlab
```

```
  Downloading reportlab-4.4.9-py3-none-any.whl.metadata (1.7 kB)
```

```
Requirement already satisfied: pillow>=9.0.0 in /usr/local/lib/python3.12/dis
```

```
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.1
```

```
Downloading reportlab-4.4.9-py3-none-any.whl (2.0 MB)
```

```
2.0/2.0 MB 21.6 MB/s eta 0:00:00
```

```
Installing collected packages: reportlab
```

```
Successfully installed reportlab-4.4.9
```

```
from scipy.stats import ttest_ind
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib.pagesizes import A4

# Run t-test (Welch's t-test)
t_stat, p_value = ttest_ind(control['revenue'], test['revenue'], equal_var=F

# Means
control_mean = control['revenue'].mean()
test_mean = test['revenue'].mean()
```

```
# Create PDF
file_path = "Revenue_AB_Test_Report.pdf" # Changed path to current director
doc = SimpleDocTemplate(file_path, pagesize=A4)
styles = getSampleStyleSheet()
story = []
```

```
story.append(Paragraph("<b>Revenue A/B Testing Report (t-test)</b>", style:  
story.append(Spacer(1, 12))
```

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```
%.colab import files  
load('Revenue_AB_Test_Report.pdf')
```

✓ The Test marketing campaign significantly outperforms the Control campaign in terms of conversion rate. The improvement is both statistically significant and practically meaningful.

Statistical significance was supported by non-overlapping confidence intervals.”

“The test showed a 3.1% absolute uplift, which is business-impactful.”

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